

AquaTrack Rwanda

Technical Progress Presentation

Task Division Plan

Group 6 - BSc. in Software Engineering Foundations Project
Presentation format: 7 minutes presentation + 3 minutes Q&A

Team Roles (current assignment)

Ange Umukundwa	Project Manager & UI/UX Lead
Merci Dieu Ndekwe	Lead Developer & Database Architect
Luigi Birasa Ntore	Research Lead & Literature Reviewer
Joseph Enitan Oke	Frontend Developer & QA Lead

Note: Merci Dieu Ndekwe and Joseph Enitan Oke have swapped roles from the original project proposal (Section 3.7). Merci now leads backend/database work; Joseph now leads frontend implementation and QA/testing.

1. Presentation Slide Allocation (7-minute talk)

Each rubric section below is assigned a single owner so grading on individual technical contribution is unambiguous. Timing is allocated to total 7 minutes; rehearse transitions between speakers.

#	Rubric Section	Pts	Owner	Time	Content to Prepare
1	Project Overview	-	Ange	1.0 min	Restate problem statement (44% NRW loss, billing transparency gap) and AquaTrack solution; link to today's technical progress.
2	Implemented Features & Tool Justification	5	Merci	1.5 min	List planned vs implemented features (anomaly detection, dashboards, SMS module); explain Django/Python choice vs alternatives (Node.js, Flask); explain anomaly-detection logic (threshold/z-score on meter deltas).
3	Database & Data Management	4	Merci	1.0 min	Present ERD (households, meter_readings, anomaly_alerts, citizen_reports, sms_notifications); explain PostgreSQL choice, indexing on household_id/timestamp, role-based access control for data security.
4	System Architecture	4	Merci + Ange	1.0 min	Show 3-tier architecture diagram (Input / Processing Engine / Output Layer); explain layered pattern rationale and scalability path (Gasabo pilot to 30 districts).
5	Code Quality & Testing	3	Joseph	1.0 min	Show coding standards (PEP8, plain HTML/CSS/JavaScript linting (ESLint for JS)); explain testing approach (pytest/Django TestCase for backend, manual usability testing); share sample test result.
6	Technical Challenges & Solutions	2	Luigi	0.5 min	Identify 1-2 challenges so far (e.g. simulating realistic meter data, feature-phone/SMS accessibility constraint); explain solution and effectiveness.
7	Feedback Integration	2	Ange	0.5 min	Summarize feedback from proposal review; state concrete changes made (e.g. scope narrowed to Gasabo pilot, SMS-first design).
8	Next Technical Steps	2	Ange	0.5 min	List upcoming milestones (data simulation script, anomaly engine, SMS integration, dashboards) with owner per Section 2 below.

Presentation Delivery (3 pts) - whole team: rehearse transitions, keep to time, consistent slide template.

Final references slide (libraries/APIs/sources cited) - Luigi compiles, whole team reviews.

2. Teamwork & Project Management Report (PDF)

Compiled by Ange (Project Manager); each member supplies evidence for their own area.

Report Section	Owner	Evidence to Supply
Project Management Tools	Ange	Screenshot of GitHub Projects board; explain how columns/labels track tasks.
Team Collaboration	Ange	Screenshots of team meetings with captions; describe standups, Git branching, design-sharing practice.
Progress Doc: Figma designs	Ange	Accessible Figma link + screenshot of citizen/WASAC dashboard wireframes.
Progress Doc: Database design	Merci	ERD screenshot + schema table; link to backend/ models.
Progress Doc: Frontend implementation	Joseph	Screenshots of HTML/CSS/JavaScript pages built so far (citizen/WASAC dashboard pages).
Progress Doc: Backend/API implementation	Merci	Screenshots of Django app structure, sample endpoint/API response.
Progress Doc: GitHub repository	Joseph	Screenshot of commit history and contributors showing collaboration.
Team Participation Sheet	Ange (Team Leader only)	Copy template, add team name, roles, participation, meeting notes.

3. Next Technical Milestones & Ownership

Milestone	Owner	Target
Data simulation script (50+ households, Gasabo)	Merci	Before next review
Anomaly detection algorithm (night-flow/spike rules)	Merci	Before next review
Django models + PostgreSQL schema migration	Merci	Before next review
Citizen dashboard (HTML/CSS/JavaScript) - usage, bill forecast	Joseph	Before next review
WASAC operator dashboard (HTML/CSS/JavaScript)	Joseph	Before next review
Unit/integration test suite (pytest)	Joseph	Before next review
Africa's Talking SMS API integration	Merci	Before next review
Usability test plan & participant recruitment (10 testers)	Ange	Before next review
Final reference list & literature cross-check	Luigi	Before next review

Reminder: grading monitors individual contributions via coding commits only. README/documentation work does not count as evidence of contribution - ensure code-owning members (Merci, Joseph) commit their own work directly.